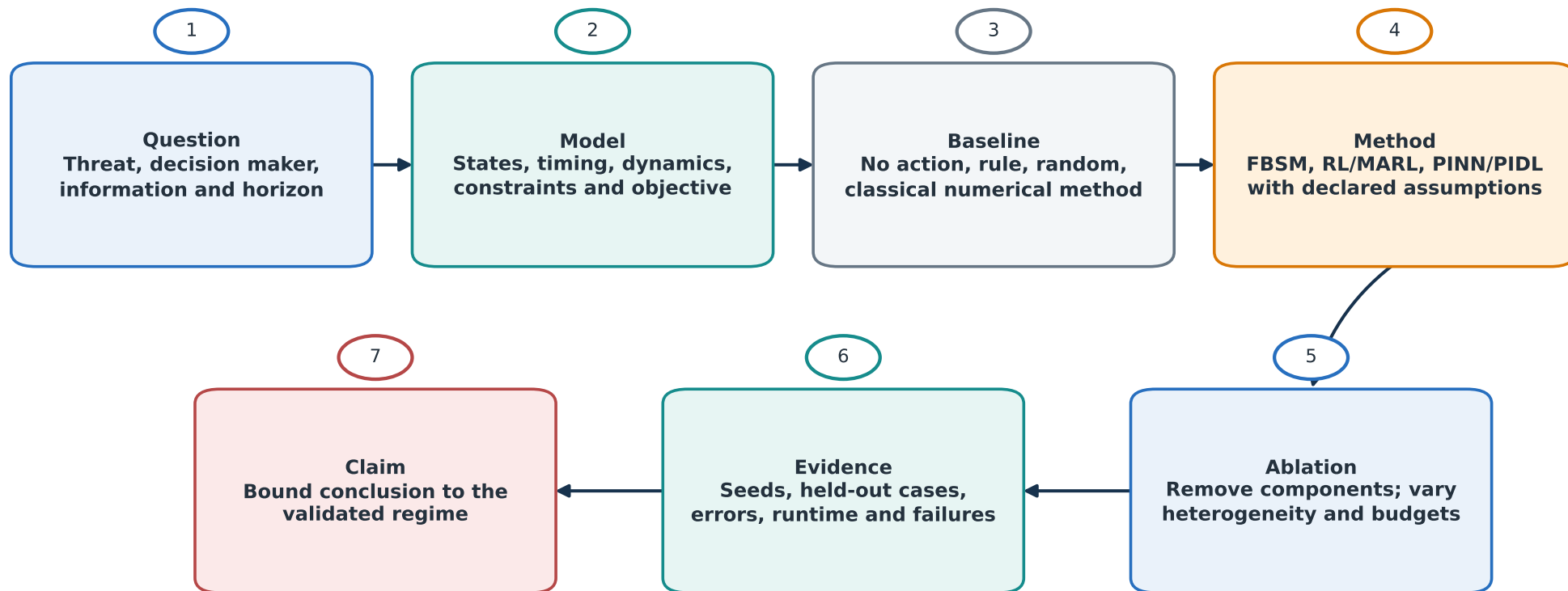


From cyber mechanism to a defensible paper claim

Each stage produces an artifact that can be inspected, rerun and challenged.



Reproducibility record: resolved parameter arrays, random seeds, graph/data provenance, hardware, software versions, training diagnostics, figure source and a machine-readable run manifest.